

Summer Bootcamp Training – DSA Questions

ARRAY

1. Find max sum of row in a matrix and discard it while printing.
2. Difference between first and last element in an array
3. Find largest element in array
4. Delete the smallest element from an array
5. Count frequency of numbers in an array
6. Sum of array (range given)
7. Leaders in an array
8. Longest consecutive increasing subarray
9. Frequency sort
10. DNF (Dutch National Flag Algorithm) Sort (0s, 1s, 2s)
11. Trapping Rainwater
12. Container With Most Water
13. Find max row in a matrix and discard it
14. Insert row and column in a matrix
15. Reverse each row in a matrix
16. Sum of boundary elements in a matrix
17. Negative of even rows and swap those rows in a matrix
18. Rotate matrix by 90°
19. Spiral traversal of matrix
20. Matrix diagonal difference
21. Rotate matrix by 180°
22. Kaden's Algorithm
23. Max Product SubArray

String

24. Count vowels

25. Create username

26. Palindrome string

27. Pattern replacement

```
str1 = ABCABCXABCXYbw
```

```
pattern = ABC
```

```
repalce_with = *
```

28. Remove duplicate characters

29. Find first non-repeating character

30. Check if two strings are anagram or not

31. Group anagrams

32. Longest Common Subsequence (LCS)

33. Longest Palindromic Subsequence

34. Longest Prefix

35. String Compression

36. Longest Substring Without Repeating Characters

37. Minimum Window Substring

38. Next Permutation of String

39. Rotate String

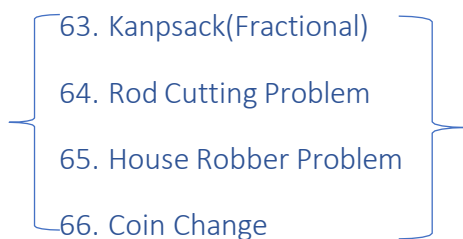
40. First Non-Repeating Character in String

41. Decode compressed string

{Extra Problems: KMP, Rabin Karp, Z algorithm, Shortest Pallindrom}

LINKEDLIST

- 42. Add two number
- 43. Convert singly to circular
- 44. Count no. of nodes
- 45. Count occurrences of given key
- 46. Delete at specific position in circular queue
- 47. Detect cycle in a linkedlist
- 48. Intersection point
- 49. Merge two sorted linkedlist
- 50. Middle node
- 51. Check linkedlist is Palindrome or not
- 52. Remove duplicates from Sorted Linkedlist
- 53. Remove every kth node
- 54. Reverse linkedlist
- 55. Update value at index in DLL
- 56. Delete Elements With Matching given Prefix
- 57. Segregate Odd Even Linkedlist
- 58. Remove Duplicate from Linkedlist
- 59. Remove node whose sum of data is Even.
- 60. Cloning of linkedlist
- 61. Flattening
- 62. Group Wise

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- 63. Knapsack(Fractional)
 - 64. Rod Cutting Problem
 - 65. House Robber Problem
 - 66. Coin Change

DP QUESTIONS

Stack & Queue

67. Menu Driven Stack Program using Linkedlist
68. Reverse String Program
69. Sum of elements in queue
70. Menu Driven Circular Queue Program
71. Stack Implementation Using Array & Linkedlist
72. Queue Implementation Using Array & Linkedlist
73. Valid Parenthesis
74. Infix,Postfix and prefix implementation & its conversion
75. Next Greater Element
76. Stock Span
77. Implementation of min stack
78. Queue using stack
79. Stack using queue
80. Sliding window maximum
81. First Negative in a window
82. Celebrity Problem
83. Largest Rectangle in histogram
84. Trapping Rainwater using stack
85. Generate Binary Number from 1 to N using Queue
86. Sort a stack using another stack
87. Delete middle element of stack
88. Implement two stack in an array
89. Reverse first k element of a queue
90. Reverse a string using stack
91. N-Stack implementation.

TREE

- 92. Inorder, postorder, preorder and levelorder Traversal
- 93. Max depth of given tree
- 94. Symmetric Tree
- 95. Invert Tree
- 96. Diameter of a tree
- 97. Check whether a tree is balanced or not
- 98. Path sum
- 99. Left-right view
- 100. Validate BST
- 101. Zig-Zag Traversal
- 102. Lowest Common Ancestor
- 103. Serialization and deserialization

Sorting and Searching

- 104. Find the number of swaps in unsorted array to sort an array.
- 105. Bubble Sort (Float Values)
- 106. Find the index of the target country in the list of countries using Binary Search.
- 107. Find the index of ticket number using linear search .
- 108. Count the number of inversion.